



ROBOTIC MANIPULATORS PROJECT

Step #1

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Application Description

The use of robotic arms is pervasive in our daily lives. This is primarily because robotic arms can replace human labor in tasks that are highly repetitive and require high precision, significantly improving production efficiency. Robotic arms can execute tasks according to predefined paths and parameters, delivering high accuracy and consistent results. In hazardous or harmful environments (such as welding or painting), robotic arms can take over tasks, ensuring the safety of human workers. From manufacturing to healthcare, logistics, and food processing, robotic arms can be customized and adjusted based on different needs. Various types of robotic arms are suited for different industrial and non-industrial applications, depending on their movement characteristics, work range, and precision requirements.

In this project, I will introduce an articulated robotic arm used for industrial automated welding of steel bars. This robotic arm is capable of moving within a designated area to weld steel bars, which typically form cubic frames, while avoiding obstacles, thereby facilitating large-scale production. The advantages of using such a robotic arm include:

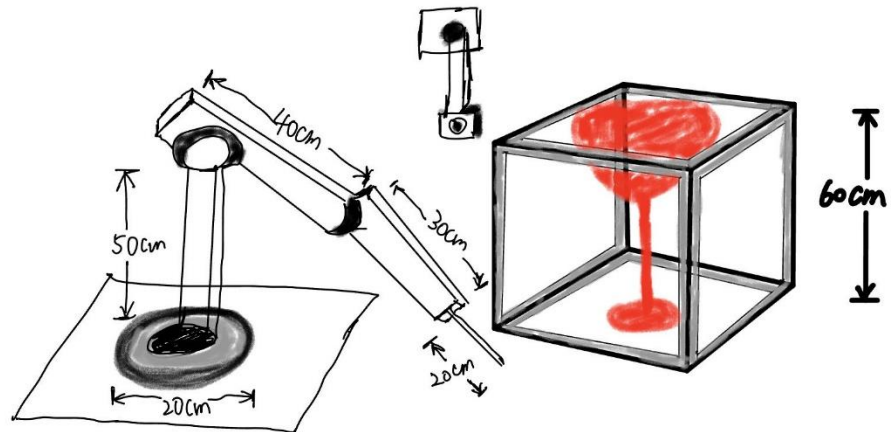
Precision and Consistency: A key advantage of using an articulated robotic arm for welding is its ability to maintain high precision and consistency. The robot can sustain a consistent welding speed, torch angle, and distance, resulting in uniform welds and reducing the likelihood of defects. Manual welding, on the other hand, is more susceptible to variations due to human factors such as fatigue.

Enhanced Safety: The welding process involves exposure to high temperatures, bright arcs, and potentially hazardous fumes. Using robots for welding reduces the need for human workers to be directly exposed to these dangers, thereby creating a safer working environment and minimizing the risk of injuries or long-term health issues.

Increased Productivity: Unlike manual welding, robots can operate continuously without the need for breaks. This is especially beneficial for large-scale projects requiring extensive welding. Robots can also work at optimal speeds, ensuring that each task is completed in the shortest time possible without compromising quality.

Flexibility: The design of the articulated robotic arm allows it to adapt to different welding tasks and reach various angles and positions within the workspace. This makes it particularly suitable for handling complex geometries, such as steel frames with intricate designs. Reprogramming the robot for different welding tasks or adjusting its path is easier than training new personnel, which further enhances its adaptability.

Long-term Cost Efficiency: Although the initial investment in robotic welding systems is high, the long-term savings can be substantial due to reduced labor costs, decreased material waste, and increased production speeds. The precision of robots also reduces the need for rework, further improving overall efficiency.



Environment Description

Since this is a welding operation, to prevent injury from sparks and the bright arc light, the entire work environment is enclosed within a large cubic black box. However, it is not entirely sealed—there is a conveyor belt at the bottom on which the steel bars that form the cubic frames are placed. Curtains are located where the conveyor interacts with the black box, allowing the steel bars to enter and exit the work area.

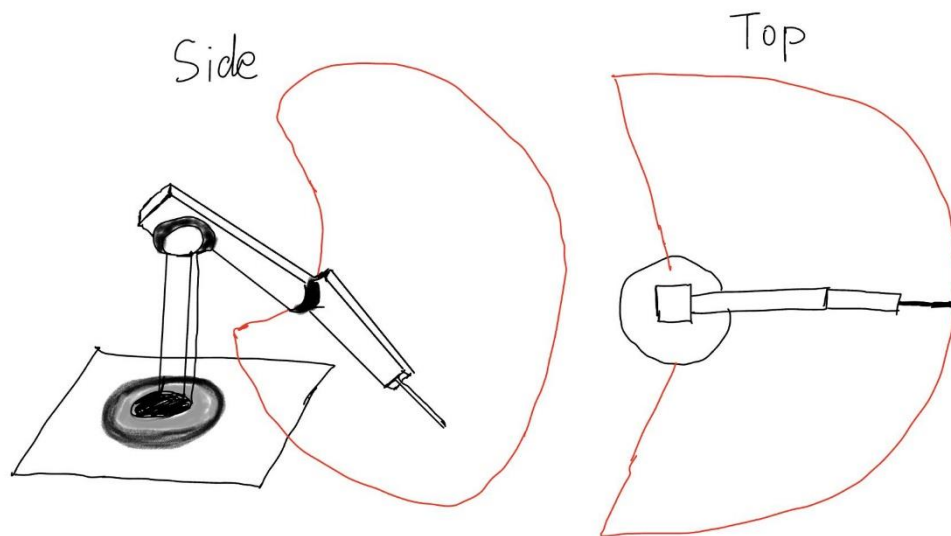
The articulated robotic arm is positioned on the floor inside the black box. Its base is a rotating disk with a diameter of 20 cm that can rotate over 180° . The body of the arm is a thick column, standing 50 cm high, with a shoulder joint at the top that allows nearly 360° rotation. The upper arm segment, measuring 40 cm, is attached to the shoulder. The elbow joint, which connects to the forearm, allows for significant angular rotation, providing further reach and flexibility. The forearm is 30 cm long and ends with a welding needle. As shown in the figure, the red-shaded area represents the workspace theoretically reachable by the robotic arm.

During the welding process, there are three obstacles:

1. **The Steel Frame Structure:** When welding the far vertices, the arm must navigate around the near portions of the frame to avoid collisions that could damage the arm or the structure.
2. **Horizontal Rod with a Camera:** A rod equipped with a camera extends from the box's edge into the central area, used for monitoring and inspecting the quality of the welds. The robotic arm must avoid this rod during operation.
3. **Conical Support Tray:** A solid cone-shaped tray, like an inverted goblet, makes contact with the top edges of the frame, providing support by holding the frame aloft for welding at the bottom vertices. This requires the arm to adjust its movements to reach the lower vertices around the solid tray.

This setup means that to weld all eight vertices, the robotic arm must avoid the

horizontally positioned rod while reaching the distant vertices, maneuvering around the solid internal tray instead of passing through the interior of the frame.



Identification of Robotic Manipulator

The selected robot for this project is an articulated robotic arm. This type of robotic arm was chosen due to its high flexibility, wide range of motion, and ability to operate in confined spaces. Unlike Cartesian robots, which move along linear axes, articulated robotic arms have multiple rotary joints that enable them to navigate around obstacles and adjust to various angles. This makes them particularly useful for welding tasks, especially when dealing with three-dimensional structures like steel frames.

The articulated robotic arm is designed with three primary segments, each with a specific length that ensures the end-effector can effectively weld each vertex of the cubic steel frame while avoiding obstacles. The detailed specifications of each segment are as follows:

- **Base and Rotating Column:** The base of the robotic arm is a 20 cm diameter rotating platform capable of 180° horizontal rotation, allowing the arm to access all sides of the workspace without needing to move its base.
- **Upper Arm:** The upper arm segment measures 40 cm in length. It is connected to the rotating column through a shoulder joint, allowing it to rotate and reach different heights. This segment is responsible for extending the arm outward towards the target welding points on the steel frame.
- **Forearm:** The forearm segment is 30 cm long, connected to the upper arm by an elbow joint that offers additional flexibility. This segment allows the arm to bend further and reach into tighter spots, such as the lower vertices of the steel frame.
- **Welding Torch (End-Effector):** The end of the forearm is equipped with a MIG welding torch and precision sensors. The total length from the base to the tip of the welding torch ensures that the robotic arm can cover the entire 60 cm

height of the cubic structure and reach around obstacles during welding.

The design of these segments provides the articulated robotic arm with a total reach of approximately 70 cm, enabling it to move around the steel frame and perform welds at all necessary points.

End-Effector Description

The end-effector of the articulated robotic arm is a MIG (Metal Inert Gas) welding torch specifically designed for industrial environments, featuring several key functionalities necessary for automated welding tasks:

- **Precision Sensors:** The welding torch is equipped with proximity sensors that monitor the distance between the welding tip and the steel surface. These sensors adjust the arc length in real-time, ensuring consistent quality, which is crucial for producing strong and uniform welds.
- **Cooling System:** Given the continuous nature of robotic welding, the torch includes an active cooling system to prevent overheating during prolonged operation. This cooling mechanism helps maintain the durability of the welding tool and ensures consistent, high-quality performance.
- **Adjustable Torch Angle:** The welding torch is mounted on a precision gripper that allows for slight adjustments to the torch's angle. This feature is particularly important when welding at various orientations or angles, such as at the corners of the steel frame. The gripper ensures that the torch maintains an optimal angle for welding, resulting in cleaner and more precise welds.
- **Auto-feed System:** The welding torch is equipped with an auto-feed system for the welding wire, providing a continuous and uninterrupted supply of material during the welding process. This system improves efficiency and reduces the need for manual intervention.

These end-effector features ensure that the articulated robotic arm meets the demands of high-precision welding in an industrial setting. The design of the welding torch, combined with the articulated movement of the arm, allows the robot to maneuver around obstacles within the workspace while maintaining high-quality welding standards.